



SOFTWARE REQUIREMENTS SPECIFICATION (SRS)
For
INNOBIZ-K STARTUP ECOSYSTEM PLATFORM
Version: 1.0



Prepared By :

Name	
1	Abenezer Endalew
2	Semeriya Seid
3	Fadil Ahmed
4	Abdi Tamene
5	Gelana Zerihun

Supervisor: Mr. Tadesse A.

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1. INTRODUCTION

1.1 Purpose

1.1.1 Objective of the Document

This Software Requirements Specification (SRS) document provides a complete, detailed, and formally structured description of the Innobiz-K Startup Ecosystem Platform. The primary objective of this document is to define, in a clear and verifiable manner, the full scope of the system's functionality, constraints, and expected behavior.

The document establishes a shared and precise understanding between all stakeholders involved in the project, including developers, system architects, project managers, evaluators, and representatives of the Ministry of Innovation and Technology. By removing ambiguity and informal interpretation, it ensures that the system is designed and implemented according to a consistent and agreed-upon specification.

In addition to guiding development, this document serves as the official reference point for system validation. Every feature implemented within the platform must be traceable to a requirement defined in this document. If a feature cannot be traced back to a requirement, it should not exist in the system. Likewise, if a requirement is defined here but not implemented, the system is incomplete.

1.1.2 Role of the Document in System Development

Within the lifecycle of the project, this document functions as a contractual and operational baseline. It is not merely descriptive; it is prescriptive. It defines what must be built, how it should behave, and under what conditions it is considered complete.

All design decisions, development tasks, and testing activities must align with the requirements specified in this document. Any changes to the system scope or behavior must be formally reviewed and documented to maintain consistency and traceability.

The SRS also plays a critical role in reducing project risk. By clearly defining expectations at an early stage, it minimizes misunderstandings, prevents scope creep, and ensures that all stakeholders are aligned before significant development resources are committed.

1.2 Scope of the System

1.2.1 System Overview

The Innobiz-K Startup Ecosystem Platform is designed as a centralized digital infrastructure that supports the lifecycle of innovation within the national startup ecosystem. It provides a unified environment where startups, investors, reviewers, and administrative bodies can interact through structured and transparent processes.

Currently, many of these interactions are managed through fragmented and disconnected tools, including spreadsheets, email communication, and online forms. These approaches create inefficiencies, including repeated data entry, inconsistent records, lack of visibility, and limited accountability. As a result, both startups and investors face unnecessary barriers when attempting to engage with one another or participate in programs.

The proposed system eliminates these inefficiencies by introducing a persistent, structured, and scalable platform. It ensures that information is captured once, stored securely, and reused across multiple processes. It also provides mechanisms for discovery, evaluation, and monitoring, thereby supporting the entire innovation lifecycle from onboarding to post-funding tracking.

1.2.2 Core System Objectives

The system is designed to achieve several core objectives that directly address the limitations of the current ecosystem.

First, it aims to standardize how information about startups and investors is collected and managed. By enforcing structured profiles, the system ensures that all relevant data is consistent, comparable, and easy to interpret.

Second, it seeks to improve visibility within the ecosystem. Startups should be discoverable based on their sector, stage, and geographic location, while investors should be identifiable based on their funding preferences and strategic focus.

Third, the system aims to enhance transparency and fairness in program management. By digitizing application and evaluation processes, it ensures that decisions are based on clearly defined criteria and that all actions are recorded and traceable.

Fourth, it introduces mechanisms for long-term monitoring and accountability. Through the tracking of milestones and performance indicators, the platform enables stakeholders to assess the real impact of funding and support programs.

1.2.3 Phased Development Approach

The system will be developed and deployed in multiple phases, each building upon the capabilities of the previous one.

The first phase establishes the core platform, focusing on profile creation, public visibility, and search functionality. This phase ensures that the platform becomes immediately useful and operational.

The second phase introduces ecosystem-level features, including automated matching between startups and investors, structured evaluation workflows, and centralized program management.

The third phase focuses on monitoring and performance tracking. It introduces tools for managing milestones, collecting performance data, and generating reports that reflect the outcomes of funding and support initiatives.

The fourth phase provides advanced operational capabilities, including facility allocation systems and automated contract generation with electronic signature support. These features enable the platform to manage not only information but also real-world resources and agreements.

1.3 Definitions and Terminology

1.3.1 Purpose of Definitions

This section defines key terms used throughout the document to ensure clarity and consistency. Given the diverse audience of this document, including both technical and non-technical stakeholders, it is essential that all terms are interpreted in a uniform manner.

1.3.2 Key Terms

A startup is defined as an early-stage business entity focused on developing innovative products or services with the intention of achieving scalable growth. Startups may exist at different stages, ranging from initial concept to growth and expansion.

An investor refers to an individual or organization that provides financial resources to startups. Investors may include venture capital firms, angel investors, grantmaking institutions, or government funding bodies.

A Key Performance Indicator (KPI) is a measurable value used to evaluate the progress or success of a startup or program. KPIs may include metrics such as revenue growth, user acquisition, job creation, or product development milestones.

A reviewer is an individual assigned to evaluate startup applications submitted to programs within the platform. Reviewers are expected to assess applications based on predefined criteria and provide objective scores and feedback.

A conflict of interest is a situation in which a reviewer's ability to make an impartial evaluation may be compromised due to personal, financial, or professional relationships with an applicant.

Designation refers to the official recognition status of a startup under the national startup proclamation. This status may affect eligibility for certain programs and benefits within the platform.

1.4 Intended Audience

1.4.1 Stakeholder Identification

This document is intended for multiple categories of stakeholders, each with distinct roles and responsibilities within the system lifecycle.

Software developers and system architects require a precise and detailed description of system behavior in order to design and implement the platform effectively. The clarity of requirements directly impacts the quality and reliability of the system.

Project managers use this document to plan development activities, allocate resources, and track progress. A well-defined SRS allows for accurate estimation and reduces the likelihood of delays or scope-related issues.

Government stakeholders and program administrators rely on the document to ensure that the system aligns with policy objectives and operational needs. The platform must support real-world processes, not just theoretical functionality.

Test engineers use the document as a basis for validation. Each requirement defined in this document must be testable, and the system must demonstrate compliance through structured testing procedures.

1.5 Document Structure

1.5.1 Organization and Navigation

This SRS is organized in a hierarchical structure that allows readers to navigate from general concepts to detailed specifications.

The first chapter provides an introduction to the system, including its purpose, scope, and context. The second chapter describes the overall system characteristics, including user types, constraints, and assumptions.

Subsequent chapters provide detailed functional and non-functional requirements, external interface descriptions, and supporting information necessary for system implementation and validation.

This structured approach ensures that the document remains both comprehensive and accessible, allowing different stakeholders to focus on the sections most relevant to their roles.

1.6 Assumptions and Dependencies

1.6.1 Operational Assumptions

The system assumes that users have access to internet-enabled devices and possess a basic level of digital literacy. It is expected that users will interact with the platform through web browsers on mobile or desktop devices.

The system also assumes that users will provide accurate and truthful information when creating profiles and submitting applications. While validation mechanisms may be implemented, the overall reliability of the system depends on user honesty.

1.6.2 External Dependencies

The functionality of the system may depend on external services and infrastructure. These include email and SMS services for notifications, as well as potential integration with national systems such as startup designation registries.

The system may also depend on hosting infrastructure provided by government or third-party providers. Performance, scalability, and security considerations must account for these dependencies.

2. OVERALL DESCRIPTION

2.1 Product Perspective

2.1.1 System Context

The Innobiz-K Startup Ecosystem Platform is designed as a centralized digital system that operates within the broader national innovation framework. It is not an isolated application; rather, it functions as a core infrastructure layer that connects multiple actors, including startups, investors, government institutions, and evaluators.

In the current environment, these actors interact through fragmented and largely manual processes. Startups submit applications through standalone forms, investors rely on informal networks to identify opportunities, and administrators manage data across spreadsheets and disconnected tools. This fragmented structure leads to inefficiencies, duplication of effort, and a lack of transparency.

The proposed system replaces these disconnected workflows with an integrated platform in which all interactions are structured, traceable, and managed within a single environment. By doing so, it introduces consistency in how information is collected, stored, and used across different processes.

2.1.2 System Architecture Overview

The platform is implemented as a web-based application built on a client-server architecture. Users interact with the system through a web interface, while the backend processes requests, enforces business rules, and manages persistent data storage.

The frontend layer is responsible for presenting information and capturing user input. It must be responsive and optimized for both mobile and desktop devices, with particular emphasis on usability in low-bandwidth environments.

The backend layer handles authentication, authorization, data processing, and system logic. It ensures that all actions performed within the system adhere to defined rules, such as access control and approval workflows.

The database layer stores all system data, including user profiles, program applications, evaluation results, and monitoring records. Data integrity and consistency are critical at this level, as the platform relies on accurate and reliable information to function effectively.

2.1.3 Relationship to External Systems

The system is designed with the expectation that it may integrate with external services and government infrastructure. These integrations may include identity verification systems, notification services such as email or SMS gateways, and official registries related to startup designation.

While such integrations are not mandatory for the initial deployment, the system architecture must remain flexible to accommodate them in future phases. This ensures that the platform can evolve without requiring fundamental redesign.

2.2 Product Functions

2.2.1 Core Functional Capabilities

At a high level, the platform provides a set of interconnected functions that support the lifecycle of startup engagement within the ecosystem.

The system enables startups and investors to create structured digital profiles that represent their identity, capabilities, and interests. These profiles serve as the foundation for all interactions within the platform.

It provides a public directory that allows users to discover startups and investors based on relevant criteria. This functionality enhances visibility and reduces reliance on informal networks.

The system also supports program management processes, including application submission, evaluation, and decision-making. These processes are designed to be transparent, structured, and efficient.

In later phases, the platform introduces advanced capabilities such as automated matching between startups and investors, post-funding monitoring through KPIs and milestones, and operational tools such as facility allocation and contract management.

2.2.2 End-to-End Workflow Integration

The system is designed to support continuous workflows rather than isolated actions. For example, a startup may begin by creating a profile, then apply to a program, undergo evaluation, receive funding, and subsequently report on progress through defined milestones.

Each of these steps is connected within the platform, allowing data to flow seamlessly from one stage to the next. This eliminates the need for repeated data entry and ensures that all relevant information is available when needed.

Similarly, investors can use the platform to discover startups, evaluate potential opportunities, and initiate communication, all within a single environment.

2.3 User Classes and Characteristics

2.3.1 Startup Users

Startup users represent founders or team members responsible for managing the startup's presence on the platform. Their primary objective is to gain visibility, access opportunities, and participate in programs.

These users may have varying levels of technical expertise. Therefore, the system must be intuitive and straightforward, minimizing complexity and guiding users through each process step by step.

2.3.2 Investor Users

Investor users include individuals and organizations that provide funding or support to startups. Their interaction with the system is primarily focused on discovering relevant startups and evaluating potential investment opportunities.

Investors typically have limited time and require efficient access to structured and reliable information. The system must therefore prioritize clarity, speed, and relevance in presenting data.

2.3.3 Administrative Users

Administrative users, including staff from the Ministry and Innobiz-K, are responsible for managing the platform's operations. Their responsibilities include approving profiles, configuring programs, assigning reviewers, and monitoring system activity.

These users require advanced access privileges and tools that allow them to oversee and control system processes effectively.

2.3.4 Reviewer Users

Reviewer users are assigned to evaluate startup applications. They are expected to provide objective assessments based on predefined criteria.

The system must support reviewers by providing clear evaluation interfaces, structured scoring mechanisms, and safeguards such as conflict-of-interest declarations.

2.3.5 Public Users

Public users are individuals who access the platform without creating an account. Their interaction is limited to browsing the public directory and viewing available information.

The system must ensure that public users can navigate and explore content easily while respecting the privacy settings defined by registered users.

2.4 Operating Environment

2.4.1 Technical Environment

The platform is designed to operate within a web-based environment accessible through modern browsers. It must support a wide range of devices, including smartphones, tablets, and desktop computers.

Given the local context, the system must perform reliably under conditions of limited internet connectivity. This requires optimization of data transfer, efficient loading of resources, and minimal reliance on heavy graphical elements.

2.4.2 User Environment

Users are expected to access the system from diverse environments, including urban and rural areas. As a result, the platform must be designed with inclusivity in mind, ensuring that it remains usable across different levels of connectivity and device capability.

2.5 Design and Implementation Constraints

2.5.1 Technical Constraints

The system must adhere to constraints related to performance, security, and scalability. It must be capable of handling increasing volumes of data and users without degradation in performance.

Security requirements include secure authentication mechanisms, role-based access control, and protection of sensitive data through encryption.

2.5.2 Regulatory Constraints

The system must comply with applicable national regulations related to data protection and digital services. This includes ensuring that user data is handled responsibly and that privacy is maintained at all times.

2.6 Assumptions and Dependencies

2.6.1 Assumptions

It is assumed that users will have access to devices capable of running modern web applications. It is also assumed that users will provide accurate information when interacting with the system.

The system assumes that administrative staff will actively manage approvals and monitor system activity to maintain data quality and integrity.

2.6.2 Dependencies

The platform may depend on external services such as notification systems for email and SMS communication. It may also rely on future integrations with government systems for verification and data synchronization.

These dependencies must be considered during system design to ensure reliability and flexibility.

3. SPECIFIC REQUIREMENTS

3.1 Functional Requirements

3.1.1 User Registration and Account Management

The system shall provide a structured user registration process that allows individuals and organizations to create accounts within the platform. During registration, users shall provide essential identity information, including name, email address, organization details, and account type. The system must validate the

authenticity and completeness of submitted information before activating the account.

After successful registration, the system shall generate a persistent account that allows users to log in securely and manage their profile information over time. Users must be able to update their details, modify profile content, and manage security credentials such as passwords. The platform shall maintain historical records of profile changes in order to ensure traceability and accountability.

The system shall also support role-based access control. Each user shall be assigned a role that determines the actions they are authorized to perform within the system. These roles include startup user, investor user, reviewer user, and administrative user.

3.1.2 Startup Profile Management

The platform shall allow startups to create and maintain comprehensive digital profiles. These profiles shall represent the startup's identity, business model, stage of development, sector focus, and operational status.

Startup users must be able to provide detailed descriptions of their product or service, the problem they aim to solve, their target market, and the technological innovation behind their solution. The system shall allow the inclusion of supporting information such as team composition, funding history, traction metrics, and external links.

The system shall ensure that startup profiles remain persistent and reusable across different processes within the platform. Once created, a startup profile shall serve as the primary reference for applications, investor discovery, and ecosystem visibility.

3.1.3 Investor Profile Management

These provide functionality for investors to create profiles that describe their investment interests and strategic priorities. Investor profiles shall include information such as investment stage preference, sector interest, funding capacity, and geographic focus.

These profiles shall enable investors to clearly communicate their expectations and investment criteria to potential startup partners. The platform shall store this

information in a structured format that allows it to be used in discovery and matching processes.

Investors must also be able to update their profiles as their investment focus evolves over time.

3.1.4 Public Startup Directory

The platform shall provide a publicly accessible directory that allows users to browse and discover registered startups. This directory shall function as a searchable database that organizes startups according to predefined attributes such as industry sector, growth stage, and location.

The system shall ensure that only approved and verified startup profiles appear in the public directory. Administrative users shall have the authority to review and approve profiles before they become visible to the public.

This feature is intended to improve visibility across the ecosystem and support collaboration between startups, investors, and other stakeholders. The Innobiz-K initiative itself focuses on supporting startups and fostering innovation within Ethiopia's ecosystem, which makes visibility and collaboration central objectives of the platform. (addisstartup.com)

3.1.5 Search and Filtering Capabilities

The system shall provide advanced search functionality that enables users to locate startups or investors based on specific criteria. These criteria may include sector, funding stage, technological focus, geographic location, and other relevant attributes. search system must support filtering mechanisms that allow users to refine results dynamically. Search results shall be presented in a structured and readable format to ensure efficient navigation.

The objective of this functionality is to enable faster discovery of relevant ecosystem participants and reduce reliance on informal connections.

3.1.6 Program Application Management

The platform shall support the management of startup support programs, incubation opportunities, and funding initiatives organized by Innobiz-K or related institutions.

Administrative users shall be able to create program listings that define eligibility criteria, application deadlines, evaluation stages, and expected outcomes.

Startup users shall be able to submit applications to available programs through the system. During submission, the platform shall automatically link the application to the startup's profile, reducing the need for repetitive data entry.

The system shall record all submitted applications and maintain a history of participation across programs.

3.1.7 Evaluation and Review Workflow

The system shall provide a structured workflow for evaluating program applications. Administrative users shall assign reviewers to applications based on predefined rules and availability.

Reviewers shall access a dedicated interface where they can evaluate applications according to specific criteria such as innovation potential, feasibility, market relevance, and team capability.

The platform shall allow reviewers to submit scores, comments, and recommendations. All evaluations shall be stored within the system and must be accessible to authorized administrative users for decision-making also include mechanisms for identifying potential conflicts of interest to preserve fairness and integrity in the evaluation process.

3.1.8 Startup–Investor Matching Mechanism

In later phases of the system, the platform shall introduce automated matching capabilities that analyze startup and investor profiles to identify potential partnerships.

The matching system shall evaluate compatibility based on sector alignment, funding stage, investment interest, and strategic objectives. The platform shall present

suggested matches to both parties, allowing them to explore potential collaboration opportunities.

This feature aims to improve efficiency in connecting startups with relevant investors and to strengthen the innovation ecosystem.

3.1.9 Monitoring and Performance Tracking

The system shall support monitoring of startups that receive support through programs or funding initiatives. Administrative users shall define performance indicators and milestones that startups are expected to achieve.

Startup users shall periodically submit updates that describe their progress toward these milestones. The system shall store and organize this information to allow administrators to evaluate the impact of support programs.

Performance tracking shall also enable the generation of reports that summarize ecosystem growth and program outcomes.

3.1.10 Facility and Resource Allocation

The platform shall provide tools for managing physical and operational resources associated with the innovation center. These may include co-working spaces, incubation facilities, and shared equipment.

Administrative users shall be able to assign resources to selected startups based on program participation and availability.

The system shall record allocation history and ensure that resources are used efficiently.

3.2 External Interface Requirements

3.2.1 User Interface Requirements

The system shall provide a modern web-based interface that supports intuitive navigation and accessibility. The design must accommodate users with varying levels of technical expertise and provide clear guidance throughout system interactions.

Interfaces shall prioritize readability, structured content presentation, and responsive design to support mobile and desktop devices.

3.2.2 Software Interfaces

The platform shall be designed to integrate with external systems where necessary. These integrations may include authentication services, government registries, and notification systems.

Such integrations must be implemented through secure and well-documented interfaces to ensure reliability and scalability.

3.2.3 Communication Interfaces

The system shall support communication channels that allow the delivery of notifications related to account activity, application status, and program updates. These notifications may be delivered through email or other messaging systems.

The communication infrastructure must ensure reliable and timely delivery of messages.

3.3 Non-Functional Requirements

3.3.1 Performance Requirements

The system shall support a large number of concurrent users without degradation in performance. Page loading times should remain within acceptable limits to ensure usability.

The platform must be optimized to function effectively even in environments where internet bandwidth is limited.

3.3.2 Security Requirements

Security is a critical requirement for the system due to the sensitive nature of user and organizational data. The system shall implement strong authentication mechanisms and enforce secure password policies.

Data transmitted between the user and the system must be protected using encryption protocols. The system must also ensure that unauthorized users cannot access restricted information.

3.3.3 Reliability and Availability

The system must maintain consistent availability to support continuous interaction among ecosystem participants. Downtime should be minimized through reliable hosting infrastructure and proper system monitoring.

Data must be backed up regularly to prevent loss of critical information.

3.3.4 Scalability

The platform shall be designed to accommodate growth in the number of users, startups, investors, and programs over time. The system architecture must support expansion without requiring major redesign.

4. SYSTEM MODELS AND REPRESENTATION

4.1 Purpose of System Modeling

4.1.1 Objective

This section provides a conceptual and logical representation of the Innobiz-K Startup Ecosystem Platform. While previous sections describe *what* the system must do, this section illustrates *how the system behaves* through structured models.

System modeling is essential for bridging the gap between requirements and implementation. It allows stakeholders to visualize interactions, data flow, and system boundaries without requiring technical implementation details.

These models support developers in designing the system architecture and assist stakeholders in understanding system behavior in a simplified but structured manner.

4.2 System Context Model

4.2.1 Overview

The Innobiz-K Platform operates within a broader ecosystem that includes multiple external and internal entities. At the highest level, the system acts as a central hub connecting startups, investors, reviewers, administrative staff, and public users.

Each of these entities interacts with the system through defined interfaces and workflows. The system itself is responsible for managing these interactions, storing data, and enforcing rules.

4.2.2 External Entities

The primary external entities interacting with the system include:

- Startups, which provide profile data, submit applications, and report progress
- Investors, who provide investment preferences and search for opportunities
- Reviewers, who evaluate applications and provide structured feedback

4.3.1 Purpose

The use case model describes how different users interact with the system to achieve specific goals. Each use case represents a complete interaction scenario between a user and the system.

4.3.2 Core Use Cases

Startup Registration and Profile Creation

A startup user creates an account, logs into the system, and completes a structured profile. The system validates the input and stores the profile for future use. The user may update the profile at any time and choose whether to make it publicly visible.

Investor Registration and Profile Management

An investor registers on the platform, provides relevant investment information, and configures visibility settings. The system stores this data and makes it available for search and matching processes.

Startup Discovery and Search

A user accesses the platform and searches for startups using filters such as sector, stage, or location. The system retrieves relevant results and presents them in a structured format.

Program Application Submission

A startup selects an available program and submits an application. The system records the submission, links it to the startup profile, and notifies administrators.

Application Evaluation

An administrator assigns reviewers to evaluate applications. Reviewers access assigned applications, submit scores and comments, and the system aggregates results for decision-making.

Matching Process

The system analyzes startup and investor profiles and generates suggested matches based on predefined criteria. Users can review these matches and initiate communication.

Post-Funding Monitoring

Startups submit periodic updates on their performance. The system records these updates and allows administrators to monitor progress against defined milestones.

4.4 Data Flow Description

4.4.1 Overview

The system processes data through a series of inputs, transformations, and outputs. Data flows between users, the system interface, backend services, and the database.

4.4.2 Data Flow Process

When a user submits information, such as a profile or application, the data is first validated at the interface level. It is then transmitted to the backend, where business rules are applied. Once validated, the data is stored in the database.

When another user requests information, such as performing a search, the system retrieves relevant data from the database, processes it according to filters or conditions, and presents the results through the interface.

This continuous flow ensures that data remains consistent, accessible, and up to date across all system components.

4.5 System Behavior and Workflow Representation

4.5.1 End-to-End Workflow

The platform supports continuous workflows that span multiple stages of interaction.

A typical workflow begins with user registration, followed by profile creation. The user may then participate in programs by submitting applications. These applications are evaluated, and decisions are recorded within the system.

If funding or support is provided, the workflow continues into the monitoring phase, where progress is tracked through milestones and performance indicators.

Each stage of this workflow is connected, allowing data to move seamlessly from one phase to the next without duplication.

4.6 Limitations of System Models

4.6.1 Scope of Representation

The models presented in this section are conceptual and do not represent the full technical implementation of the system. They are intended to provide clarity and understanding rather than detailed design specifications.

More detailed representations, such as database schemas and technical architecture diagrams, may be included in later sections or appendices.

5. USE CASE SCENARIOS

5.1 Purpose

This section provides detailed descriptions of key system interactions from the perspective of users. Unlike the abstract use case model, these scenarios describe step-by-step flows of actions and system responses.

5.2 Use Case: Startup Profile Creation

5.2.1 Description

This use case describes the process through which a startup creates and manages its profile within the platform.

5.2.2 Actors

Primary Actor: Startup User

System Actor: Innobiz-K Platform

5.2.3 Flow of Events

The process begins when the startup user accesses the platform and selects the option to create an account. The user provides required registration details and successfully logs into the system.

After authentication, the user navigates to the profile creation section and begins entering startup information. The system guides the user through structured input fields, ensuring that all required data is provided.

Once the profile is completed, the user reviews the information and submits it. The system validates the input and stores the profile. The user may then choose to make the profile publicly visible, subject to administrative approval.

5.3 Use Case: Program Application

5.3.1 Description

This use case describes how a startup applies for a program within the platform.

5.3.2 Flow of Events

The startup user browses available programs and selects one that matches their interest. The system displays detailed information about the program, including eligibility criteria and deadlines.

The user initiates the application process and completes the required fields. The system allows the user to save progress and return later if needed.

Upon submission, the system records the application and notifies the relevant administrative users. The startup receives confirmation that the application has been successfully submitted.

5.4 Use Case: Application Evaluation

5.4.1 Description

This use case describes how reviewers evaluate submitted applications.

5.4.2 Flow of Events

An administrator assigns applications to reviewers. Reviewers log into the system and access their assigned tasks.

Each reviewer examines the application and provides scores based on predefined criteria. The system records these scores and aggregates results.

After all evaluations are complete, administrative users review the results and make a final decision.

6. DATA REQUIREMENTS

6.1 Purpose

The purpose of this chapter is to define the structure, storage, and constraints of the data required by the Innobiz-K Startup Ecosystem Platform. Proper data definition ensures consistency, reliability, and traceability throughout all system processes.

6.2 Data Entities

The primary data entities of the system are described below:

6.2.1 Startup Profile

- **Attributes:** Startup ID, Name, Sector, Stage, Team Members, Product/Service Description, Funding History, Traction Metrics, Website/Links.
- **Constraints:** Each startup must have a unique ID; mandatory fields include Name, Sector, and Stage.
- **Relationships:** Linked to Applications, Funding Records, Facility Allocations.

6.2.2 Investor Profile

- **Attributes:** Investor ID, Name, Investment Preferences, Sector Focus, Funding Capacity, Geographic Focus, Contact Info.
- **Constraints:** Unique ID; mandatory fields include Name and Investment Preferences.
- **Relationships:** Linked to Matching Records, Communication Logs.

6.2.3 Program Application

- **Attributes:** Application ID, Startup ID, Program ID, Submission Date, Status, Scores, Reviewer Comments.
- **Constraints:** Each application must be linked to one startup and one program.
- **Relationships:** Tied to Startup Profiles and Reviewer Evaluations.

6.2.4 Reviewer Evaluation

- **Attributes:** Evaluation ID, Reviewer ID, Application ID, Score, Comments, Date of Review.
- **Constraints:** Reviewers cannot evaluate applications where conflicts of interest exist.
- **Relationships:** Associated with Applications and Programs.

6.2.5 Facility Allocation

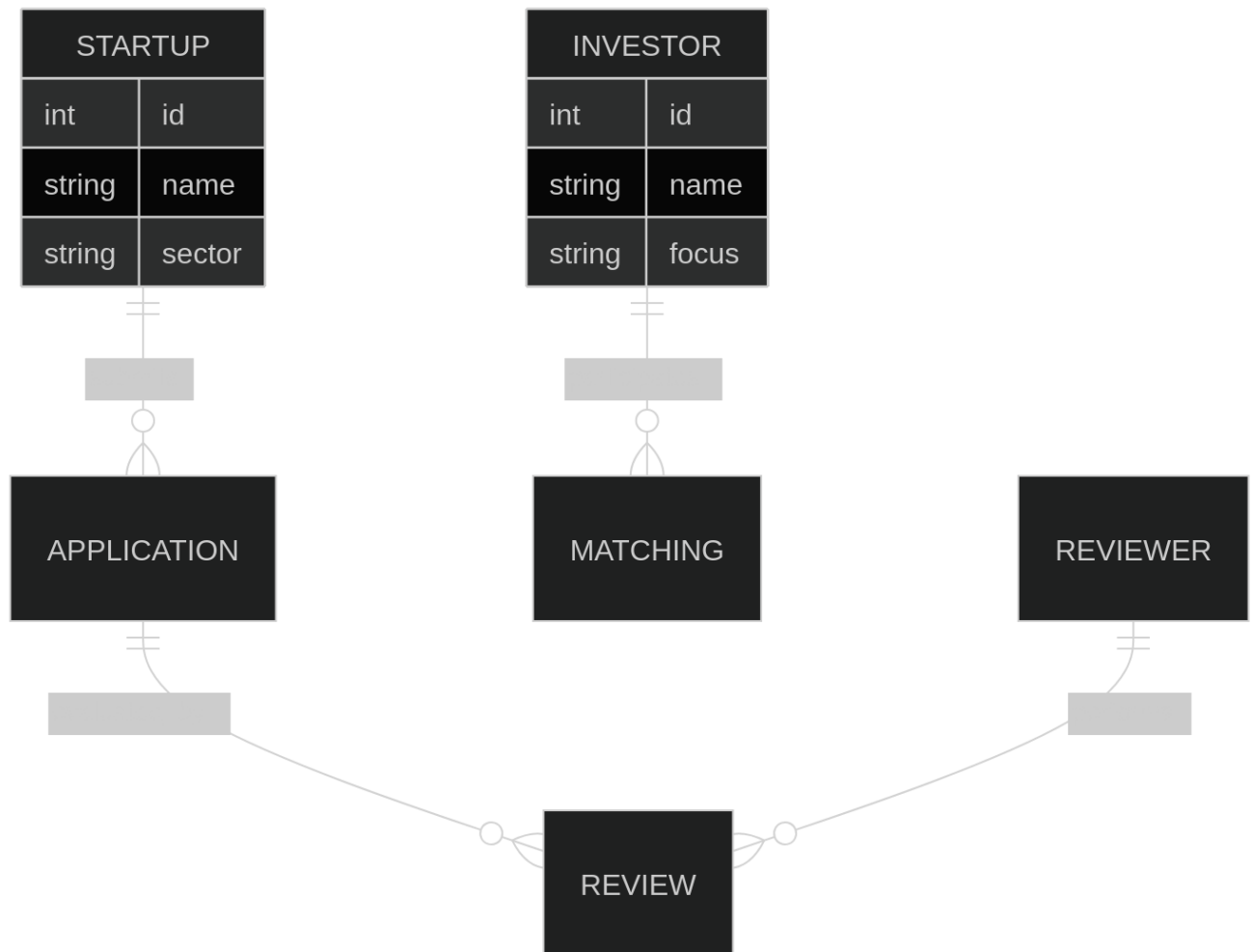
- **Attributes:** Allocation ID, Startup ID, Facility ID, Start Date, End Date, Status.
- **Constraints:** Each allocation must reference a valid startup and facility; overlapping allocations must be prevented.
- **Relationships:** Linked to Startups and Facilities.

6.2.6 Communication Log

- **Attributes:** Log ID, Sender ID, Recipient ID, Message, Timestamp, Type (Email/SMS/Platform).
- **Constraints:** Messages must be linked to valid users.
- **Relationships:** Tied to Users and Notifications.

6.3 Entity Relationship Diagram (ERD)

The ER diagram represents the database structure of the system. It includes key entities such as users, startups, investors, programs, applications, and their relationships. The system follows a relational database design where foreign keys establish connections between entities.



6.4 Data Storage and Management

6.4.1 Database Design

The platform shall use a relational database to store structured information. Primary keys, foreign keys, and indexes shall be implemented to ensure data integrity and query performance. The database shall include tables corresponding to all major entities described above.

6.4.2 Data Security

Sensitive information such as user credentials, contact info, and funding records shall be encrypted at rest and during transmission. Role-based access control shall restrict access to data based on user privileges.

6.4.3 Backup and Recovery

The system shall implement automated daily backups to prevent data loss. Recovery procedures must allow restoration to the last consistent backup within one hour.

6.5 Data Validation Rules

The system shall enforce input validation on all forms. For example:

- Email addresses must follow standard format.
- Numeric fields (e.g., funding amounts) must be non-negative.
- Mandatory fields must be completed before submission.
- Duplicate records (e.g., duplicate startup names in the same sector) shall be prevented.

7. APPENDICES AND SUPPORTING INFORMATION

7.1 Acronyms and Abbreviations

Acronym	Meaning
KPI	Key Performance Indicator
SRS	Software Requirements Specification
UI	User Interface
UX	User Experience

7.2 User Interface Mockups

Figure 1: **Admin Dashboard**

This interface represents the administrative dashboard used to manage system operations. It provides an overview of platform activity, including user management, program administration, and system monitoring. The dashboard enables administrators to approve profiles, configure programs, assign reviewers, and oversee overall platform performance.

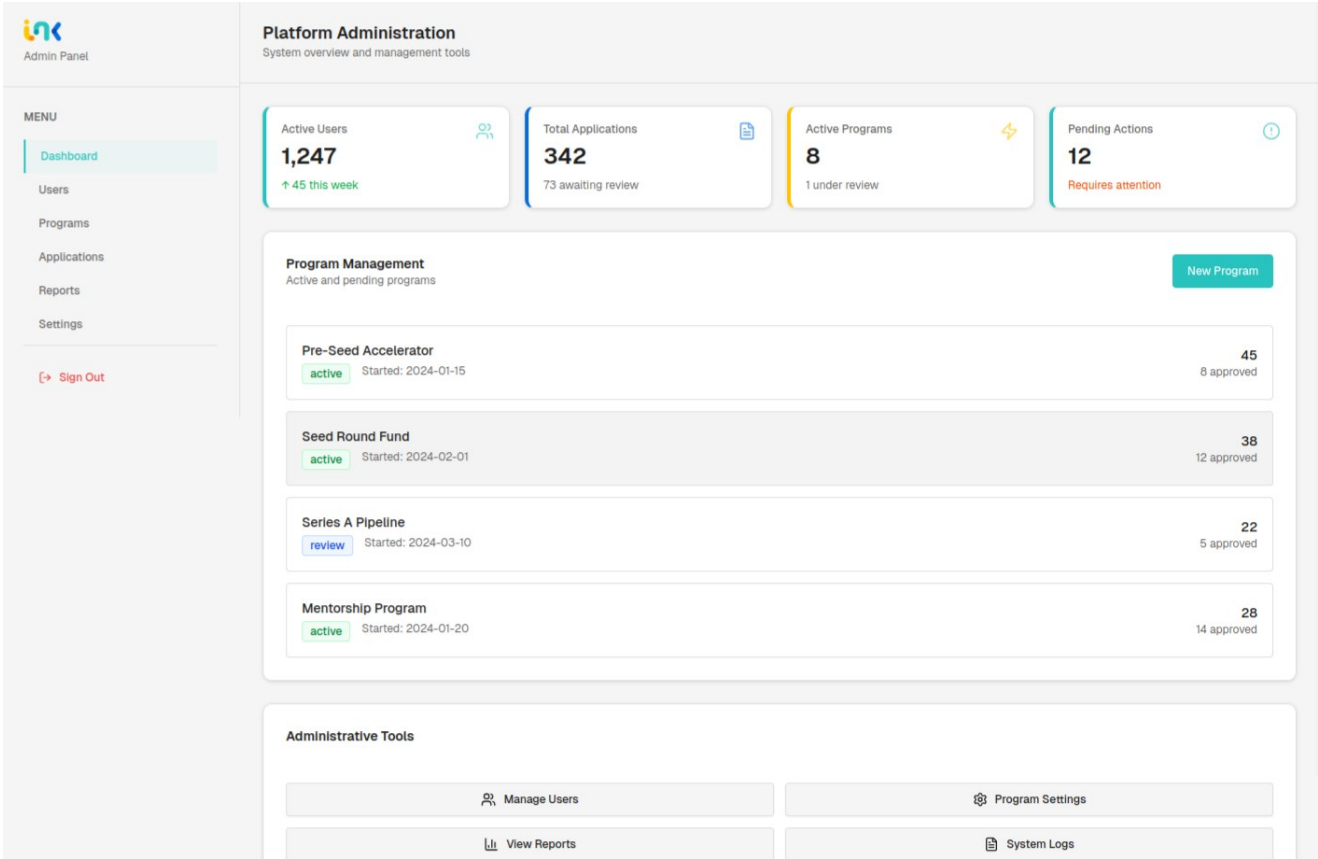


Figure 2: Reviewer Dashboard

This interface represents the reviewer dashboard. It allows reviewers to access assigned applications, evaluate submissions based on predefined criteria, and provide scores and feedback. The dashboard is designed to streamline the evaluation process and ensure consistency and transparency in decision-making.

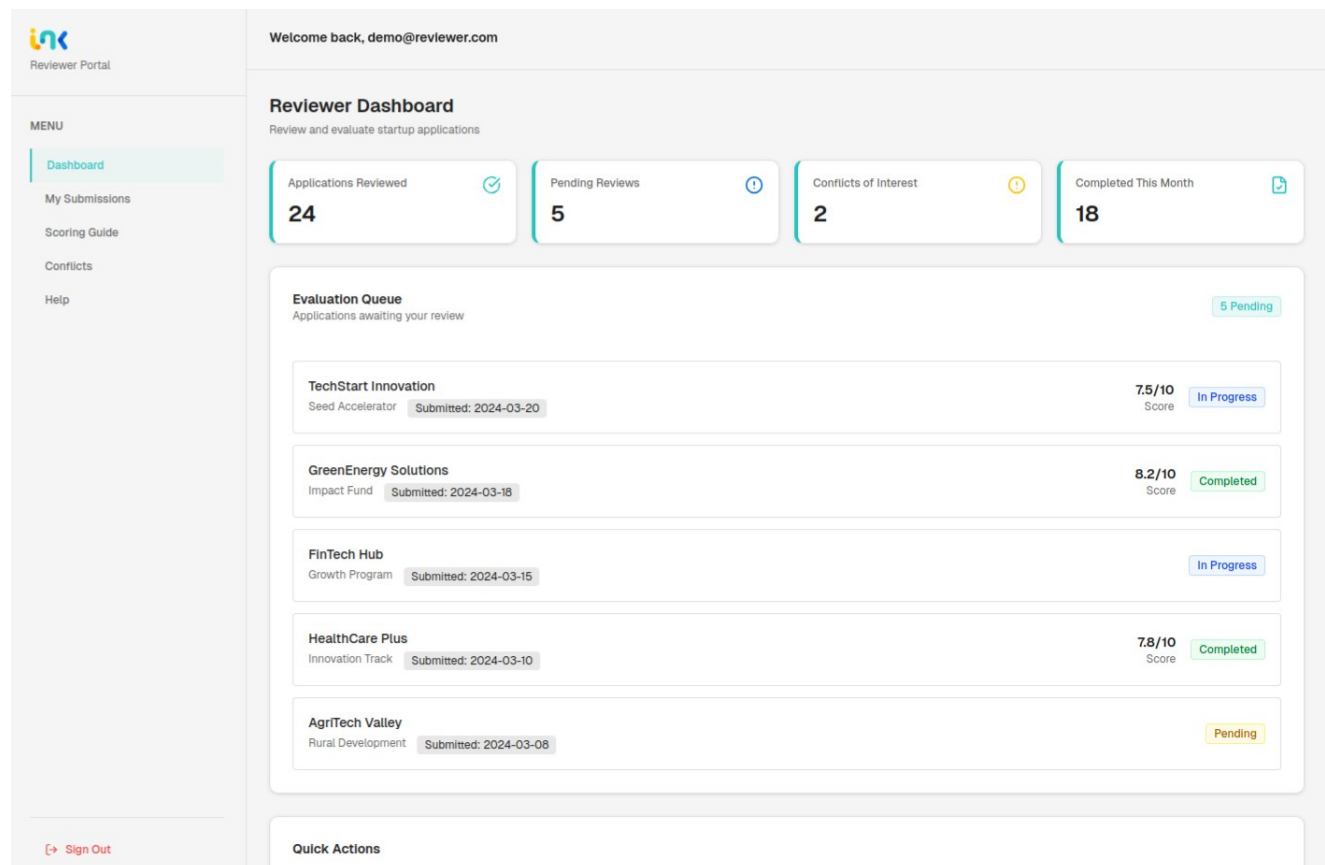


Figure 3: Investor Dashboard

This interface represents the dashboard for investor users. It displays important investment-related insights, including portfolio size, capital invested, returns, and available opportunities. Visual charts illustrate portfolio distribution and sector allocation, while the interface also highlights potential investment opportunities that match the investor’s preferences.

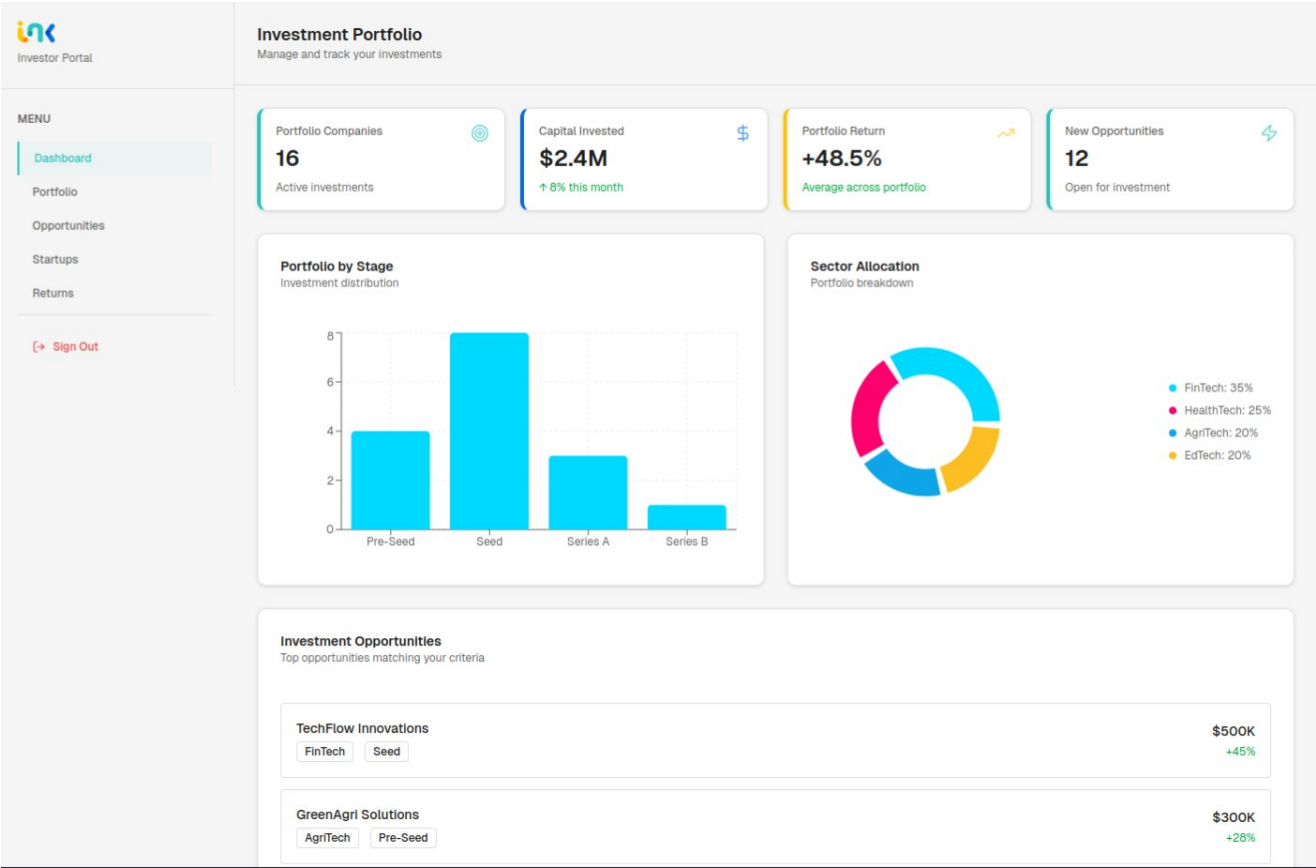
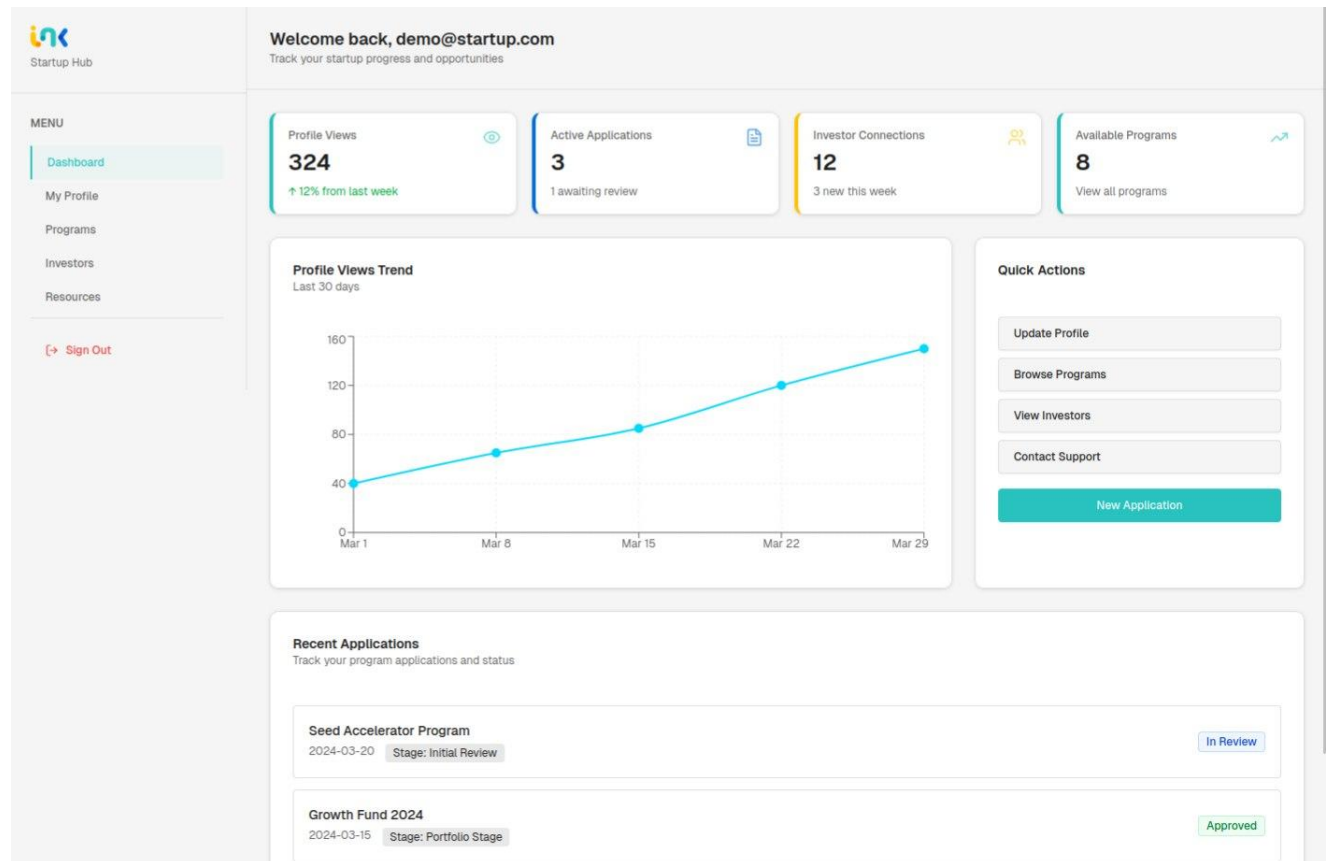


Figure 4: Startup Dashboard

This interface represents the dashboard for startup users. It provides an overview of key activities, including profile views, active applications, investor connections, and available programs. The dashboard also includes a trend visualization for performance monitoring, recent application status updates, and quick action buttons for common tasks such as updating profiles and applying to programs.



7.3 Reference Documents

1. Innobiz-K Startup Ecosystem Guidelines (2026)
2. National Startup Proclamation, Ethiopia
3. Technical Design Documents for Web Platform Architecture
4. User Experience Guidelines for Government Portals

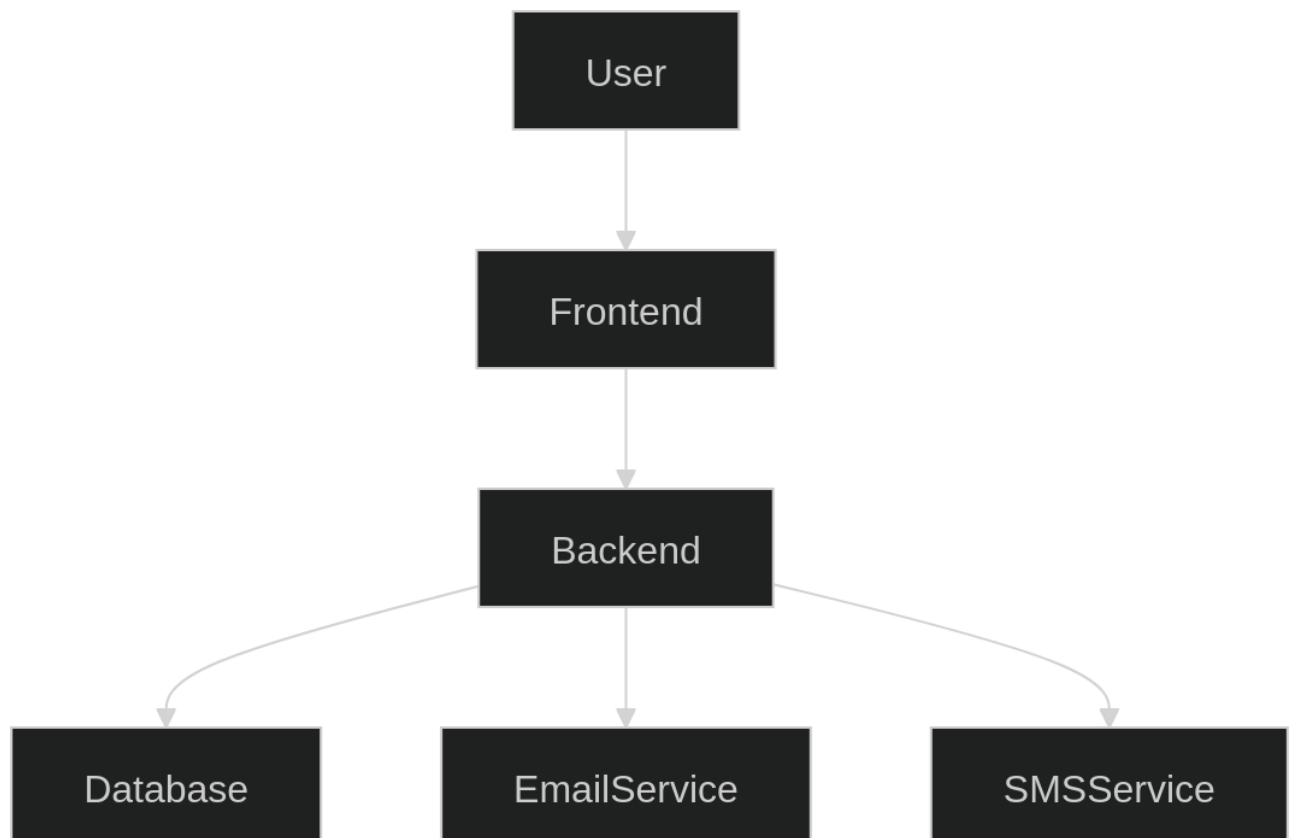
7.4 Architecture Diagrams

7.4.1 System Architecture Overview

- Web-based client-server architecture
- Frontend: Responsive web application
- Backend: API-driven application layer with role-based access
- Database: Relational DBMS with encryption and auditing

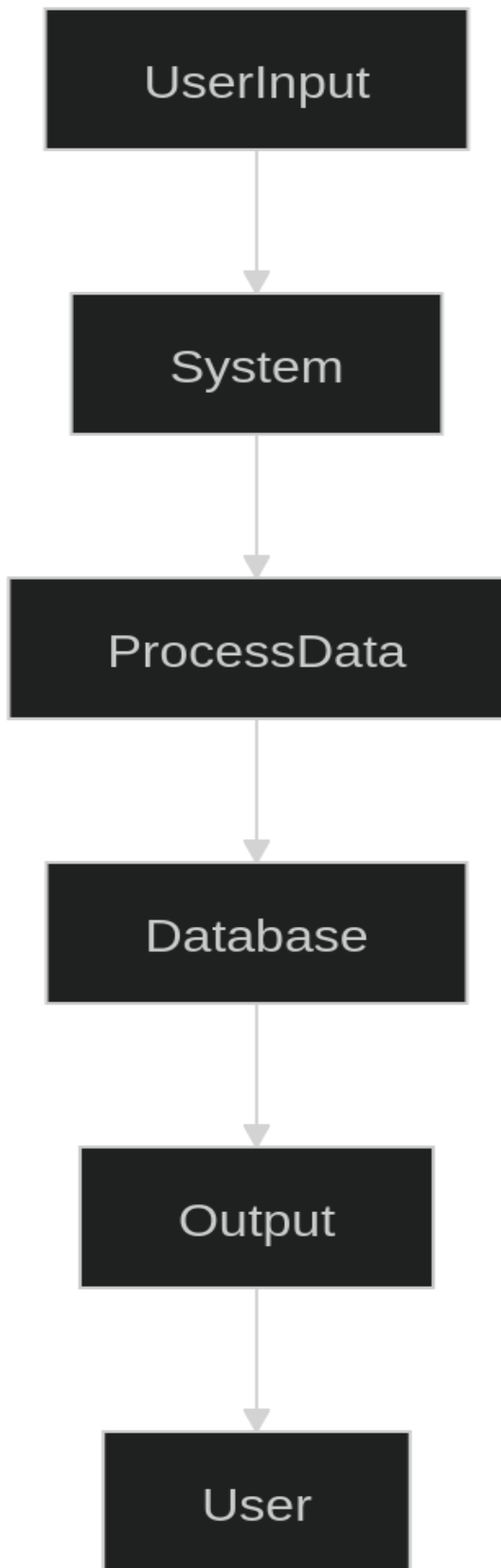
7.4.2 System Architecture Diagram

The system architecture diagram illustrates the overall structure of the Innobiz-K Startup Ecosystem Platform. The system follows a client-server architecture, where users interact with the platform through a web-based frontend interface. The frontend communicates with the backend server, which handles business logic, authentication, and application processing. The backend is connected to a relational database that stores user profiles, applications, and system data. External services such as email and notification systems are integrated to support communication and system functionality.



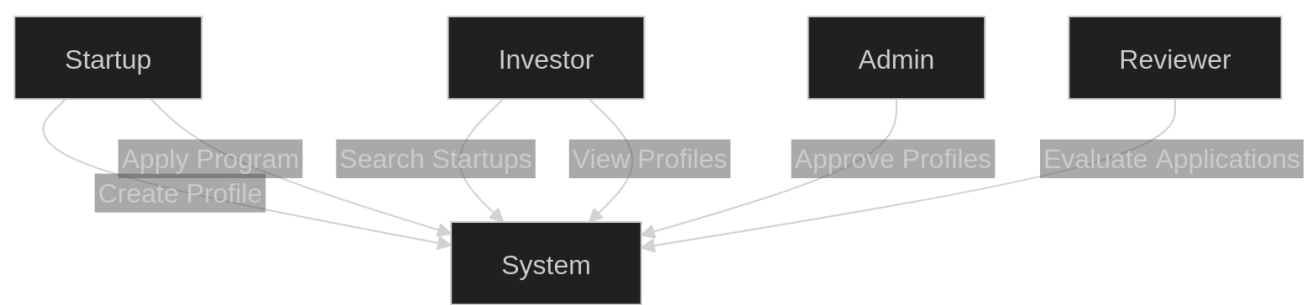
7.4.3 Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how data moves through the system from user input to final output. Users such as startups, investors, reviewers, and administrators provide input through the frontend interface. The data is validated and sent to the backend for processing, where business logic and authentication are applied. The processed data is then stored in the database. When required, the system retrieves data from the database and presents it to users in the form of reports, application status updates, or notifications. This process ensures efficient data handling, consistency, and real-time interaction within the system.



7.4.4 Use Case Diagram

The use case diagram illustrates the interaction between system users (Startup, Investor, Reviewer, and Administrator) and the platform. It presents the main functionalities such as user registration, profile management, program application, application evaluation, and system monitoring.



7.5 Assumptions and Constraints

- Users have internet-enabled devices.
- Government staff actively manage approvals and allocations.
- Notifications depend on external email/SMS services.
- The system must scale to at least 5,000 concurrent users in Phase 1 and grow for future phases.

7.6 Revision History

Version	Date	Author	Changes
1.0	23-Mar-26	Innobiz-K Team	Initial Draft
1.1	23-Mar-26	Innobiz-K Team	Expanded Chapters & Models

